Aralytics: project status

Feature list, checklist, gaps and recommendations for the Ponce AP10 Learning Hub (aralytics.cleolyn.dev). Status as of 7 November 2026, main branch at commit 3edce46.

At a glance

Area	Status
Core learning (lessons, quizzes, tasks, gradebook, analytics)	Built and tested
Accounts (Clerk sign-in, enrollment codes, admin-approved instructors)	Built and live
File storage (sharded Cloudflare D1, signed links, quotas)	Live; first real upload still to verify
Automated tests	12 server suites, 109 client tests, all passing
Production hardening (Clerk production keys, CI, rate limits, backups)	Not done yet

1. What we have

Accounts and access

• Sign-in with Google or email through Clerk; verified email required for every private API.

• Three roles from the database only: Admin, Teacher (instructor), Student. Clerk metadata is ignored.

• Instructor applications: a new account applies at /instructor/apply; the admin approves or returns it with a reason.

• Admin dashboard (/admin): waiting, approved and returned applications, current instructors, storage usage.

• admin:grant command to create the first admin (dry run by default).

• Students join a class with an adviser-issued code; the adviser approves or rejects the request.

Teaching (instructors)

• Classes page with enrollment codes and approval queue.

• Lessons hub: upload files or add HTTPS links, organized by competency, quarter and ILAW step.

• Quiz builder with six formats: multiple choice, true/false, identification, matching, situational, PISA-style data analysis. Drafts saved on the device.

• Performance tasks: create, collect submissions, grade with score and comment.

• Gradebook, proficiency profiling, item analysis (difficulty and discrimination indices).

• Remediation queue with Approve / Modify / Override actions; support plans page.

• Proctoring audit trail (tab switches and window blur, with a flag limit per quiz).

• Exports: XLSX master gradebook and DepEd monitoring PDF report.

Learning (students)

• Dashboard, lessons with reading logs, offline lesson downloads (PWA) that sync when back online.

• Proctored quiz runner: answers and timer kept on the device, no repeat submissions, survives connection loss.

• Activities: submit performance tasks with an upload progress bar.

• Progress page and suggested practice / interventions.

Files

• My files page for every instructor and student.

• Chunked uploads (1 MiB parts, automatic retry) into 8 free Cloudflare D1 databases (about 3.5 GB in total).

• Private, short-lived signed download links; permission checks per role.

• Limits: instructors 15 MB per file and 1 GB each; students 10 MB per file and 50 MB each; site cap 3.5 GB.

• Daily cleanup of unfinished uploads (03:17 UTC). Admin can see usage and delete a person’s files.

Platform

• npm workspaces monorepo: React 18 + Vite client, Express 5 API.

• Database adapter for SQLite (local), Turso (production) and MySQL.

• Deployed as one Cloudflare Worker on the free plan; deploy with npm run deploy:cloudflare.

• Tests: server flow tests on a throwaway SQLite database, Vitest + Testing Library UI tests.

• Docs: README, enrollment-and-auth, cloudflare-deployment, brand guide, sample outputs.

2. Checklist

[x] done · [~] partly done · [ ] missing

Accounts and administration

[x]	**Clerk sign-in, verified email, database roles**	Done
[x]	**Admin-approved instructor applications**	Done
[x]	**Admin account set up in production** The site owner's account holds the Admin role.	Done
[ ]	**Admin can revoke instructor access or change a role** Today a role can only be raised (approval or admin:grant).	Missing
[ ]	**Admin user management: list, suspend or delete accounts** No account-deletion flow exists. It must also delete the person's files.	Missing
[ ]	**Admin audit log** Reviews record who and when, but there is no general log of admin actions.	Missing
[ ]	**Email notifications** No email on application approved or returned, enrollment decisions, or new grades.	Missing
[~]	**Clerk production instance** Site uses Clerk test keys (pk_test). Development badges are hidden in code, but that change is not deployed yet.	Partial

Teaching and learning

[x]	**Lessons, quizzes (6 formats), performance tasks, grading**	Done
[x]	**Gradebook, analytics, item analysis, remediation queue**	Done
[x]	**XLSX and PDF exports**	Done
[x]	**Offline lessons (PWA)**	Done
[ ]	**Edit or delete a published quiz** The API has create and read only.	Missing
[ ]	**Edit or delete a performance task** The API has create, submit and grade only.	Missing
[~]	**Students see only their adviser's lessons** Downloads are enforced; the lesson list itself is not filtered.	Partial
[ ]	**A student in more than one class** One enrollment per student by design.	Missing
[ ]	**Announcements or messages between teacher and class**	Missing
[ ]	**Filipino-language interface** The interface is English only.	Missing

Files

[x]	**Chunked uploads, signed links, quotas, cleanup, admin storage tab**	Done
[~]	**Verified on real Cloudflare** Covered by tests with a stand-in; the first real upload and download still need a manual check.	Partial
[~]	**Access revoked immediately** Download links stay valid up to 6 hours after access is removed.	Partial
[ ]	**More than 3.5 GB** Limited by the free plan (10 D1 databases, 500 MB each). R2 needs a card on file.	Missing

Engineering and operations

[x]	**Automated tests for server and UI** 12 server suites and 109 UI tests passing.	**Done**
[ ]	**Continuous integration (GitHub Actions)** Tests only run when someone runs them locally.	**Missing**
[ ]	**Lint and formatting (ESLint, Prettier)** No lint or format scripts.	**Missing**
[ ]	**End-to-end browser tests** No Playwright or Cypress tests of the full sign-in to quiz flow.	**Missing**
[ ]	**Rate limiting and security headers** No rate limiter or helmet-style headers on the API.	**Missing**
[ ]	**Automated database backups** Only manual backups were taken during the reset.	**Missing**
[~]	**Monitoring** Cloudflare observability logs are on; no alerts or error tracking.	**Partial**
[ ]	**Staging environment** Changes go from local straight to production.	**Missing**
[~]	**Repository tidiness** Unrelated files at the root: Parking_Validation_Research_Activity.pdf, output/; skills-lock.json is untracked.	**Partial**
[~]	**Stale test section** redesign_flow live-HTTP section still uses the old upload and login calls.	**Partial**

3. Recommendations

Ordered by priority. Every item fits the free Cloudflare plan.

#	Recommendation	Why	Effort
1	Deploy the hidden Clerk badges, then test one real upload and download from the admin account.	Confirms the new file storage works on real Cloudflare.	Minutes
2	Move Clerk to a production instance for aralytics.cleolyn.dev.	Removes development mode for good; test keys have usage limits and are not meant for real users.	1–2 hours (DNS records in Cloudflare)
3	Add GitHub Actions: run npm test and the build on every push and pull request.	Catches broken changes before they reach production.	About 1 hour
4	Add rate limiting on sign-in, enrollment and upload routes, plus security headers.	Protects the free-plan request budget and the accounts.	Half a day
5	Quiz and performance-task editing and deleting.	Teachers cannot fix a mistake after publishing.	1–2 days
6	Admin user management: revoke instructor access, suspend and delete accounts (with file cleanup).	Needed to handle staff leaving or wrong approvals.	1–2 days
7	Email notifications for applications, enrollment decisions and grades (free tier of a mail service).	People currently have to keep checking the site.	1 day
8	Scheduled Turso backups (daily export) and a written restore drill.	Protects grades and records against mistakes.	Half a day
9	End-to-end browser tests for the main journeys and a staging Worker.	Tests the real sign-in and real storage before production.	2–3 days
10	Tidy the repository: remove unrelated root files, add ESLint and Prettier, fix the stale test section.	Keeps the project easy to work on.	Half a day
11	Filter the student lesson list by adviser; consider a Filipino interface option.	Clearer for students; better fit for the school.	1–3 days